David Feldman

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Education

University of Illinois Urbana-Champaign Urbana, IL
B.S. Mathematics and Computer Science - GPA: 4.0 *Expected Grad May 2028*

- State Farm CS Scholarship: 1 of 2 from a cohort of 2,500+ recognized for exceptional academic promise
- Societies: James Scholar Honors Program, Quant @ Illinois (Software Division), SIAM, ACM SIGma, SIGPLAN
- Coursework: Computational Geometry, Algorithms and Models of Computation, Computer Architecture, Data Structures

Research

Illinois Mathematics Lab Jan 2026 — May 2026
UIUC Math Department *Urbana, IL*

- Researched computational approaches to homotopy groups of spheres, fundamental in higher-dimensional topology
- Built optimized Rust and Python tools that computed further into the EHP spectral sequence than the algorithm's author
- Produced the most detailed visualizations of the sequence to date; presented findings in a research poster (repo)

Selected Projects

Real-Time Audio Visualization Suite Nov 2025

- Manual Rust implementation of iterative Cooley-Tukey FFT; real-time spectrogram with synchronized WAV playback
- Lock-free SPSC ring buffer for cross-thread audio streaming; memory-mapped WAV parser with zero-copy sample reinterpretation and lazy bidirectional sliding-window cache

High Performance Particle Physics Engine Mar 2025

- Multithreaded physics engine simulating kinematics, gravity, and collisions for 50,000+ particles at 60 FPS in Rust
- Lock-free parallelization with atomics, grid spatial partitioning and sweep for collision detection; Verlet integration

Eulerian Fluid Simulation Aug 2025

- Interactive real-time 2D CFD simulator with incompressible flow, obstacle interaction, and multiple visualization modes
- MAC grid velocity field with backward advection sampling, divergence-free projection for pressure solving in Rust

Constraint Satisfaction Puzzle Solver Jul 2025

- Flow Free game clone and CSP solver with constraint propagation and forward checking heuristics in Rust
- Automated puzzle collection pipeline via Python web scraper with 1500+ extracted levels

Automatic Differentiation Engine Jul 2025

- Rust clone of Karpathy's micrograd; automatic differentiation engine, topological dag sort for arbitrary expression gradient
- Used to implement backpropagation fundamentals; usable as library dependency for neural network training

End-to-end Mechanical Keyboard Dec 2023

- Fully custom printed PCB with QMK/C, community group buy and open-sourced case hardware and firmware

Skills

- **Languages**: Rust, Python, C/C++, OCaml, Typst, LaTeX
- **Tools**: Git, Cargo, rayon, cargo-flamegraph, PyTorch, NumPy
- **Concepts**: Numerical Methods, Optimization, Computational Geometry, Computational Linear Algebra, Concurrency

Experience

Full-Stack Developer Sep 2023 — May 2025
Tech Team, Los Altos Hacks *Los Altos, CA*

- Developed software infrastructure for the world's largest high school hackathon for two years in a row
- Coordinated feature development on multi-team React/TypeScript projects through Git workflows
- Led the judging system sub-team, implementing custom pairwise voting to streamline in-hackathon voting